

Task 1: Backup and Disaster Recovery Plan

Prepared for: Ubuntu Innovations (Pty) Ltd

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1. Purpose and Scope

This Backup and Disaster Recovery (BDR) Plan defines the procedures, schedules, and responsibilities required to protect Ubuntu Innovations (Pty) Ltd against data loss, system failure, and service disruption. The plan covers all 25 employees across five departments and applies to every IT asset deployed in the Cape Town office.

2. Recovery Objectives

Objective	Target	Definition
Recovery Time Objective (RTO)	< 4 hours	Maximum acceptable time to restore IT systems and resume normal business operations following a failure or disaster.
Recovery Point Objective (RPO)	< 24 hours	Maximum acceptable amount of data loss measured in time. No more than 24 hours of business data may be lost in any recovery scenario.

3. Backup Schedule

The following three-tier backup schedule ensures data is protected at multiple points in time, satisfying both the RTO and RPO targets above.

Frequency	Backup Type	Time	Retention	Storage Target	Scope
Daily	Incremental	23:00 SAST	7 days rolling	NAS (RAID-5) + Cloud	Changed files since last backup — all departments
Weekly	Full	Friday 22:00	4 weeks	NAS (RAID-5) + Cloud	Complete backup of all company data and system configs
Monthly	Archive	Last Friday	12 months	Cloud offsite only	Full snapshot archived for compliance and audit purposes
Continuous	Cloud Sync	Real-time	Real-time	Cloud offsite only	Critical files (Finance, HR) replicated continuously

4. Backup Storage Architecture

Ubuntu Innovations uses a two-location (on-premises + cloud) storage strategy to ensure data is always recoverable even in the event of physical office damage.

Layer	Technology	Specification	Role
Primary (On-site)	NAS Device	Synology DS423+ — 8TB RAID-5	First backup target for daily and weekly jobs; RAID-5 provides single-disk fault tolerance
Secondary (Offsite)	Cloud Storage	Microsoft Azure Backup / OneDrive for Business	Encrypted offsite copy; protects against fire, theft, or physical disaster at the office
Tertiary (Archive)	Cloud Cold Storage	Azure Blob Storage (Cool tier)	Low-cost long-term archive for monthly snapshots retained for 12 months

5. Data Classification and Backup Priority

Priority	Department	Data Type	Backup Frequency	Justification
Critical	Finance	Payroll, invoices, SARS records	Daily + Continuous	Regulatory compliance; data loss has legal consequences
Critical	Human Resources	Employee records, contracts, DocuSign	Daily + Continuous	Privacy Act compliance; sensitive personal information
High	Executive Mgmt	Strategic docs, Power BI reports	Daily	Business decision-making; confidential information
High	Software Dev	Source code, GitHub repos, Docker images	Daily	Core business IP; loss would halt development operations
Medium	Sales & Marketing	CRM data, campaigns, analytics	Daily	Revenue-impacting; cloud tools (HubSpot) have their own backups
Standard	All Staff	Microsoft 365 email and files	Daily	M365 includes built-in redundancy; supplemented by local backup

6. Disaster Recovery Procedures

The following step-by-step procedures must be followed when invoking this plan. The IT Support Specialist is responsible for executing all recovery steps.

6.1 Minor Data Loss (Single File or Folder)

Step	Action	Detail
1	Identify the loss	Confirm with the affected user which file(s) are missing or corrupted and when they were last accessible.
2	Access NAS backup	Log into the Synology NAS management console and navigate to the shared folder snapshot for the relevant date.
3	Restore the file	Copy the required version from the NAS snapshot back to the live shared folder or the user's device.
4	Verify integrity	Ask the user to open and confirm the restored file is complete and correct.
5	Log the incident	Record the incident, cause, and resolution in the IT troubleshooting log.

6.2 Major System Failure (Server or NAS Failure)

Step	Action	Detail
1	Declare the incident	Notify management immediately. Activate this BDR plan and assign the IT Support Specialist as incident lead.
2	Isolate failed hardware	Disconnect the failed device from the network to prevent data corruption spreading to other systems.
3	Assess extent of damage	Determine which systems, data, and users are affected. Document scope before attempting recovery.
4	Restore from NAS backup	If NAS is intact, restore the most recent full backup to a replacement or repaired server within the 4-hour RTO window.
5	Restore from cloud	If the NAS is also affected, initiate a restore from the most recent cloud backup via Microsoft Azure Backup.
6	Test restored systems	Verify all critical services (file sharing, printing, internet, M365) are operational before returning users to the network.
7	Notify all staff	Communicate the resolution and any data loss window (RPO) to department heads and management.
8	Post-incident review	Document lessons learned and update this BDR plan within 5 business days of recovery.

6.3 Full Office Disaster (Fire, Flood, or Theft)

Step	Action	Detail
1	Ensure staff safety	Physical safety of all 25 employees takes priority over all IT recovery actions.
2	Contact management	Notify the CEO and department heads immediately. Confirm the office is inaccessible or destroyed.

3	Activate remote work	Direct all staff to work remotely via Microsoft 365 (cloud-hosted). VPN access to be enabled by IT Specialist.
4	Initiate cloud restore	Begin full restoration of all data from Azure cloud backup to temporary or replacement hardware.
5	Procure replacement HW	Contact hardware suppliers to procure replacement desktops, NAS, and networking equipment under warranty.
6	Rebuild infrastructure	Reconfigure the network, reinstall all software, and restore data from cloud within the 4-hour RTO target.
7	Test and validate	Confirm all systems, user accounts, shared folders, and applications are functional before declaring recovery complete.
8	Report and review	File an insurance claim if applicable. Conduct a full BDR review and update this document accordingly.

7. Backup Testing and Verification

Backups that are never tested cannot be trusted. The following testing schedule must be strictly observed.

Test Type	Frequency	Procedure	Pass Criteria
File restore test	Weekly	Restore a random sample of 5 files from the previous night's incremental backup to a test folder.	All 5 files restored correctly with no corruption within 15 minutes.
Full restore test	Monthly	Restore the most recent full backup to an isolated test environment (VirtualBox VM).	All shared folders, user accounts, and applications accessible within 4 hours.
Cloud restore test	Quarterly	Initiate a partial restore from Azure cloud backup to verify offsite data integrity.	Selected data restored successfully; hash verification passes.
DR simulation	Annually	Simulate a full office failure scenario: shut down on-premises systems and restore from cloud only.	Full business operations restored from cloud within RTO of 4 hours.

8. Roles and Responsibilities

Role	Responsibility
IT Support Specialist	Execute all backup and recovery procedures; maintain and test the BDR plan; report outcomes to management.
CEO / Executive Management	Approve this plan; declare a disaster event; authorise emergency hardware procurement.
Department Heads	Report data loss or system failures immediately to the IT Support Specialist; communicate recovery status to their teams.

All Staff	Report suspected data loss or system issues promptly; do not attempt self-recovery without IT approval.
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9. Plan Review and Maintenance

- This plan must be reviewed and updated every 6 months or immediately following any disaster recovery event.
- Any changes to IT infrastructure (new hardware, cloud providers, or staff growth) must be reflected in an updated version of this document.
- The IT Support Specialist is responsible for version control and ensuring all department heads have access to the current plan.

Version	Date	Author	Changes
1.0	June 2025	IT Support Specialist	Initial document